Assignment 4

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1 Effect of Reward Shaping

Figure 1 compares sparse and shaped reward structures. The sparse reward agent receives +50 only upon reaching the goal, while the shaped reward agent receives -1 per step, -5 for obstacle collisions, and +50 for goal achievement.

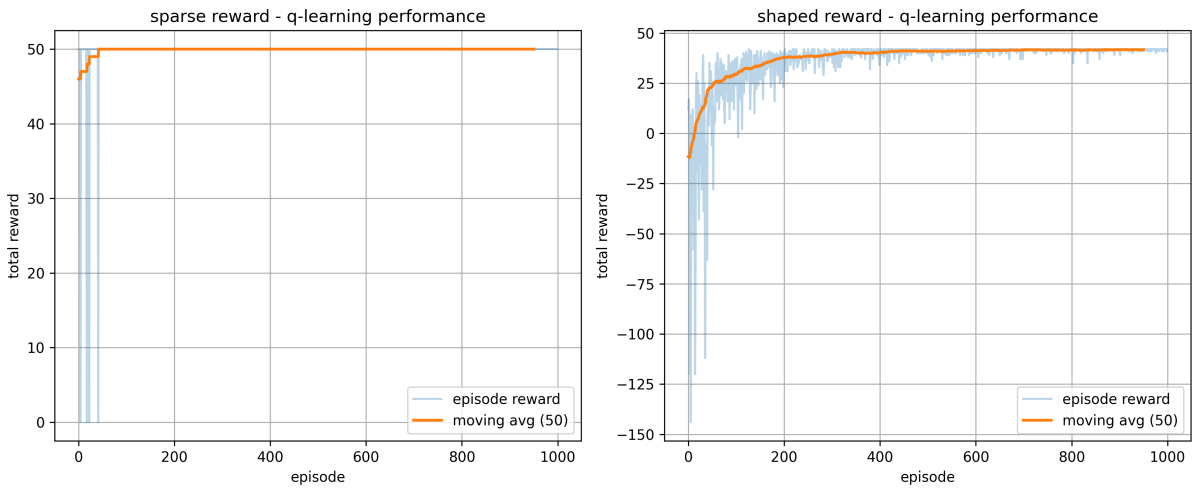

Figure 1: Sparse reward (left) vs shaped reward (right) learning performance

The sparse reward agent shows erratic learning in early episodes, with most episodes yielding zero reward. Learning is slow and unstable, taking approximately 150 episodes before consistent goal achievement. Without intermediate feedback, the agent must randomly explore until accidentally discovering the goal.

The shaped reward agent demonstrates much smoother and faster convergence. Starting from highly negative rewards (around -100), the agent steadily improves as step penalties guide it toward shorter paths. Shaped rewards provide feedback at every step, helping the agent understand progress toward the goal. This results in faster convergence (around 100 episodes) and significantly lower variance.

Figure 2 shows distance-based reward shaping using Manhattan distance, where reward = -0.1 $\times$ distance. This approach shows the fastest learning, converging around episode 80 by creating a gradient that naturally guides exploration toward the goal.

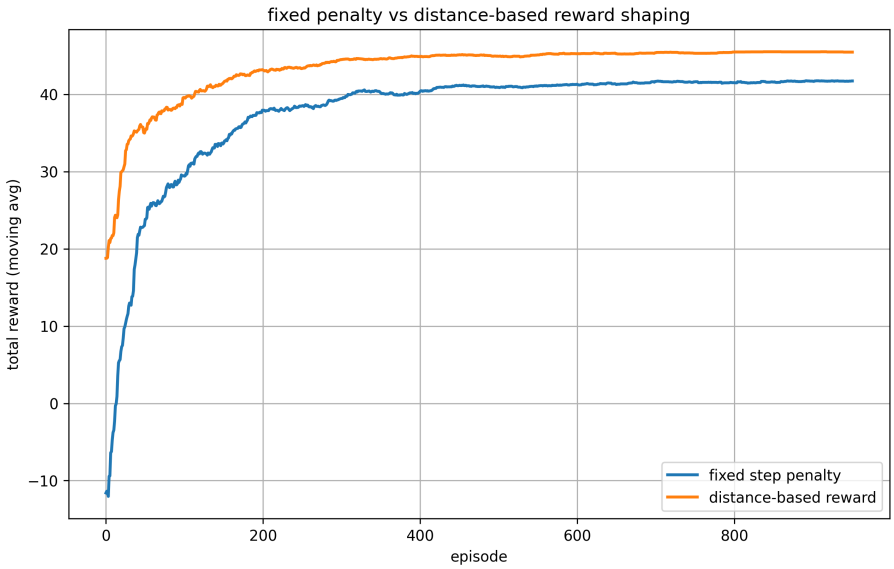

Figure 2: Fixed step penalty vs distance-based reward shaping

2 Effect of Stochastic Wind

Wind dynamics introduce stochasticity where the same action can produce different outcomes. Columns 2-3 have wind strength 1, and columns 4-5 have wind strength 2, pushing the agent upward with probability 0.8. This creates three key challenges: the environment becomes non-deterministic requiring multiple experiences to learn expected values, wind can push the agent into obstacles disrupting intended actions, and stochasticity increases variance in episode rewards visible as noisy traces in all plots. Shaped rewards help mitigate this variance by providing consistent feedback even when wind disrupts movement.

3 Effect of Adaptive Wind

Figure 3 compares fixed wind (p=0.8) against adaptive wind where probability changes: p=0.6 (episodes 1-300), p=0.8 (episodes 301-700), and p=0.9 (episodes 701-1000).

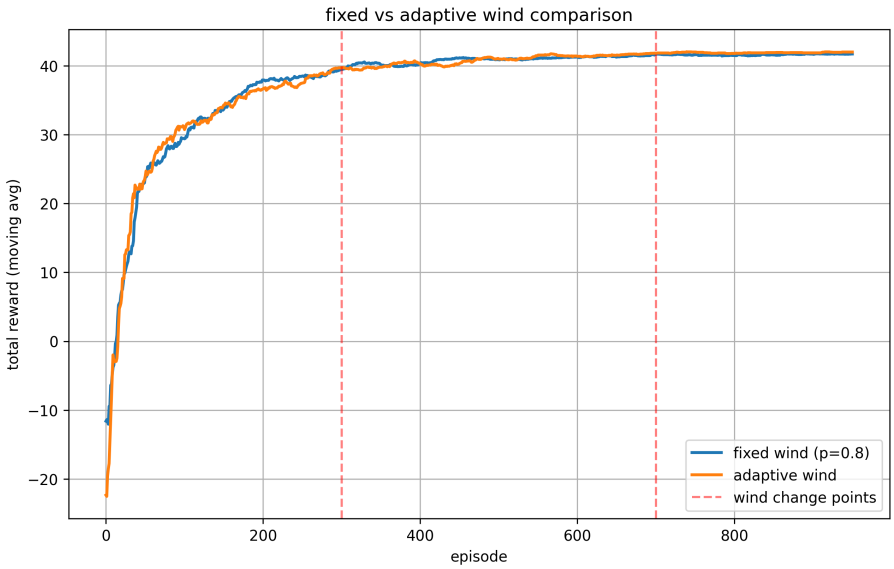

Figure 3: Fixed wind vs adaptive wind under non-stationary dynamics

The adaptive wind creates a non-stationary environment where dynamics change over time. At transition points (red dashed lines), slight performance dips occur as the learned policy becomes less optimal. When wind probability increases from 0.6 to 0.8 at episode 300, and again to 0.9 at episode 700, minor fluctuations appear. However, Q-learning adapts relatively quickly by continuing to update based on new experiences. Final performance converges to similar levels, showing Q-learning can handle moderate non-stationarity.

4 Impact of Energy Constraint

Figure 4 shows performance with a limited energy budget of 20 units, where each move costs 1 unit and wind pushes cost an additional 1 unit.

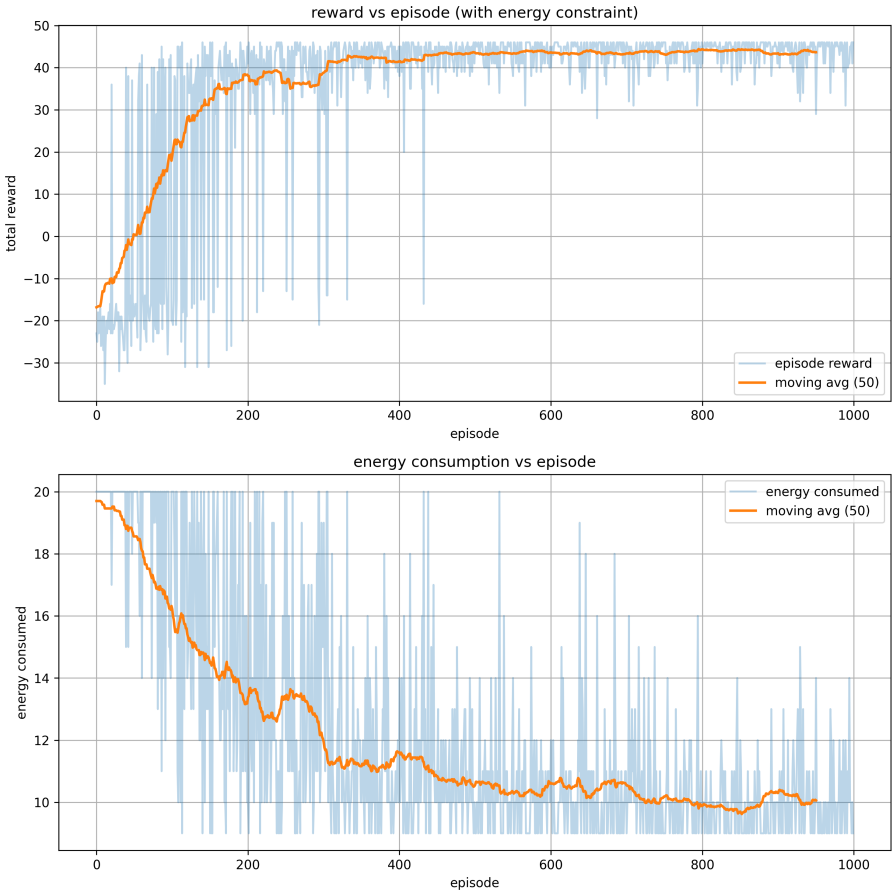

Figure 4: Reward (top) and energy consumption (bottom) under energy constraints

The energy constraint introduces a trade-off between path length and success probability. Early episodes frequently exhaust energy before reaching the goal, resulting in very low rewards. The agent must quickly discover direct paths or face termination.

Interestingly, the energy-constrained agent achieves higher average rewards (around 44) compared to unconstrained shaped reward (around 40). This occurs because energy constraints force the discovery of more efficient paths, reducing unnecessary exploration. The energy consumption plot shows learning progression clearly: initial episodes consume nearly all 20 units, while trained agents stabilize around 10-11 units, indicating an efficient path that accounts for wind effects. The energy constraint acts as implicit reward shaping, penalizing long paths more severely than simple step penalties.